

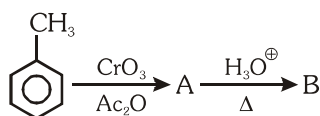
CARBONYL COMPOUNDS



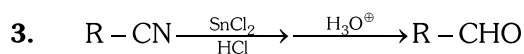
Reducing agent is

- (1) Red P and HI (2) Pd + BaSO₄ / H₂
(3) LiAlH₄ / H⁺ (4) Zn-Hg / conc. HCl

2. Final product of the following reaction is



- (1) Benzylalcohol (2) Benzoic acid
(3) Benzaldehyde (4) Toluene



Above reaction is

- (1) Mendius reaction (2) Stephen's reaction
(3) Rosenmund reaction (4) Perkin reaction

4. Which of the following reaction do not form benzaldehyde

- (1) Etard reaction
(2) Gatterman-Koch reaction
(3) Reimer Tieman's reaction
(4) Rosenmund's reaction

5. Which of the following is used as a reducing agent in clemmensen reduction

- (1) Zn-Hg/HCl (2) NH₂-NH₂/OH⁻
(3) Red P and HF (4) LiAlH₄/H⁺

6. Which of the following has maximum Boiling point

- (1) Methoxy ethane (2) n-Butane
(3) Propanal (4) Acetone

7. Arrange the following in decreasing order of boiling point :-

- CH₃CHO (I), CH₃CH₂OH (II)
CH₃OCH₃ (III), CH₃CH₂CH₃ (IV)
(1) II > III > I > IV (2) IV > III > I > II
(3) II > I > III > IV (4) I > II > III > IV

8. Which of the following do not reduce Fehling solution

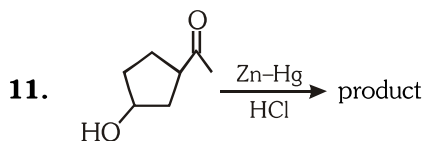
- (1) CH₃CHO (2) Ph-CHO
(3) HCOOH (4) HCHO

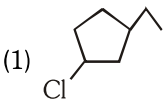
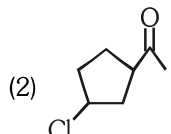
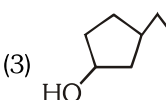
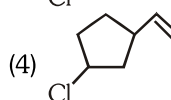
9. Which of the following do not give aldol reaction with dil. NaOH

- (1) CH₃CHO (2) CD₃CHO
(3) CH₃COCH₃ (4) PhCHO

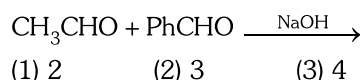
10. Arrange the following in decreasing order of reactivity towards nucleophilic addition

- HCHO (I), CH₃CHO (II)
CH₃COCH₃ (III), PhCOCH₃ (IV)
(1) II > IV > III > I (2) I > II > III > IV
(3) IV > III > II > I (4) II > III > IV > I



- (1)  (2) 
(3)  (4) 

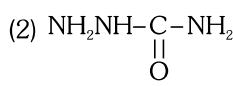
12. How many product will be obtained in the following reaction



13. Which of the following contains most acidic hydrogen

- (1) CH₃CHO (2) CH₃COCH₂COCH₃
(3) CH₃COCH₂COOCH₃ (4) CH₃COCH₂CONH₂

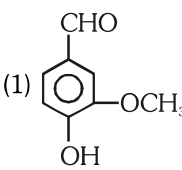
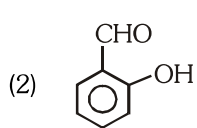
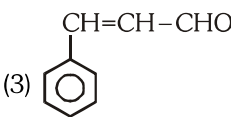
14. CH₃CHO reacts most readily with

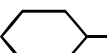
- (1) NH₂-NH₂ (2) 
(3) Ph-NH-NH₂ (4) H₂N-OH

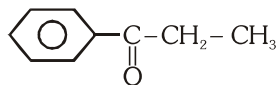
15. Which of the following do not give cannizaro reaction

- (1) CCl₃CHO (2) HCHO
(3) PhCHO (4) (CH₃)₃CCHO

16. Which have pleasant fragrances

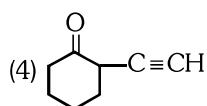
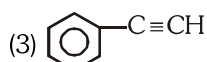
- (1)  (Vanillin)
(2)  (Salicylaldehyde)
(3)  (Cinnamaldehyde)
(4) All of them

17. IUPAC name of  and



respectively are –

- (1) Cyclohexane carbaldehyde and 1-Phenyl propanone
 (2) Cyclohexyl methanal and Ethyl phenyl ketone
 (3) Cyclohexane methanal and Propionoxy benzene
 (4) Cyclohexane methanal and 1-Phenyl propanone
18. Which gives aldehyde on hydration by H_2SO_4 and HgSO_4
- (1) $\text{CH}_3 - \text{C} \equiv \text{CH}$
 (2) $\text{CH}_3 - \text{C} \equiv \text{C} - \text{CH}_2 - \text{CH}_3$



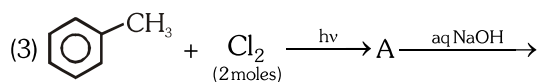
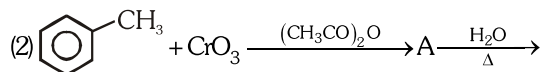
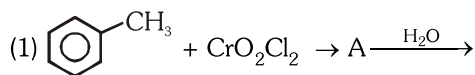
19. $\text{CH}_3 - \text{C} \equiv \text{N} \xrightarrow[\text{+HCl}]{\text{SnCl}_2} \text{A} \xrightarrow[\text{H}_2\text{O}]{\text{H}^+} \text{B} \xrightarrow[\text{Ni}]{\text{H}_2} \text{C}$
 'C' is

- (1) $\text{CH}_3 - \text{CH}_2 - \text{OH}$ (2) $\text{CH}_3 - \text{COOH}$
 (3) $\text{CH}_3 - \text{CHO}$ (4) $\text{CH}_3 - \text{CH}_2 - \text{NH}_2$

20. $\text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_2 - \text{CH}_2 - \text{CN} \xrightarrow[2. \text{H}_2\text{O}]{1. \text{DIBAL-H}} ?$

- (1) $\text{CH}_3(\text{CH}_2)_4\text{CN}$
 (2) $\text{CH}_3(\text{CH}_2)_5\text{NH}_2$
 (3) $\text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_2 - \text{CH}_2 - \text{CHO}$
 (4) $\text{CH}_3(\text{CH}_2)_3 - \text{CH}_2 - \text{CHO}$

21. Which gives Benzaldehyde



- (4) All of them

22.  + $\text{CO} + \text{HCl} \xrightarrow[\text{AlCl}_3]{\text{Anhyd.}} \text{C}_6\text{H}_5\text{CHO}$

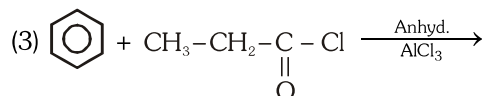
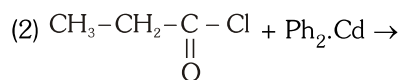
above reaction is called as

- (1) Gatterman aldehyde synthesis
 (2) Gatterman - Koch reaction
 (3) Koch reaction
 (4) Etard reaction

23. The Best reagent to convert Allyl alcohol in propenal

- (1) $\text{O}_3/\text{H}_2\text{O}/\text{Zn}$ (2) H^+/KMnO_4
 (3) PCC (4) H_2O_2

24. In which reaction ethyl phenyl ketone is obtained



- (4) All of them

25. Give reactivity order towards HCN

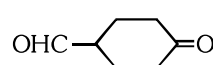
- (I) $\text{Ph} - \text{COCH}_3$
 (II) $\text{CH}_3 - \text{CO} - \text{CH}_3$
 (III) $\text{CH}_3 - \text{CHO}$
 (IV) $\text{Ph} - \text{COCH}_2 - \text{CH}_3$

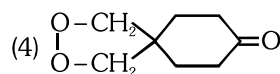
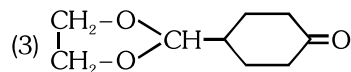
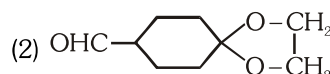
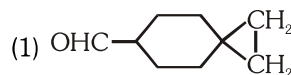
- (1) $\text{IV} > \text{I} > \text{II} > \text{III}$

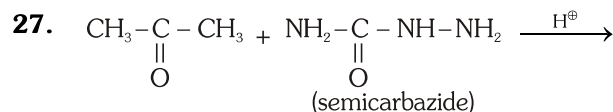
- (2) $\text{III} > \text{II} > \text{I} > \text{IV}$

- (3) $\text{II} > \text{I} > \text{IV} > \text{III}$

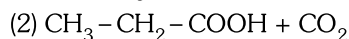
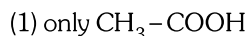
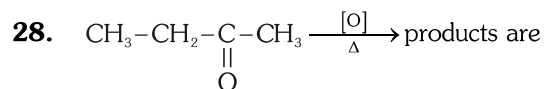
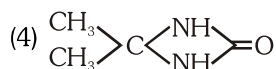
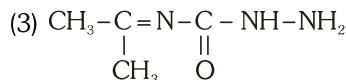
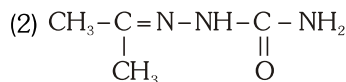
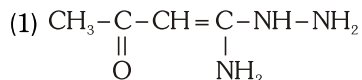
- (4) $\text{III} > \text{II} > \text{IV} > \text{I}$

26.  + $\begin{matrix} \text{CH}_2 - \text{OH} \\ | \\ \text{CH}_2 - \text{OH} \end{matrix} \xrightarrow{\text{H}^+} \text{A 'A' is}$
 (1 mole)

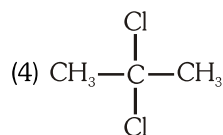
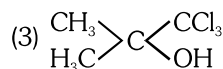
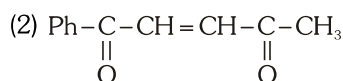
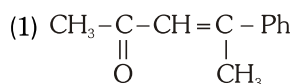
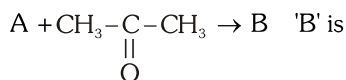
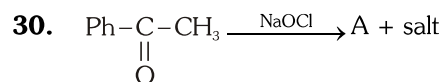
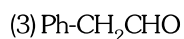
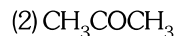
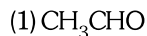




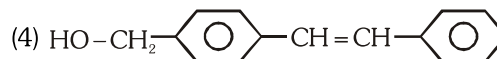
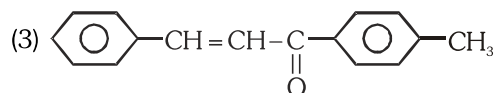
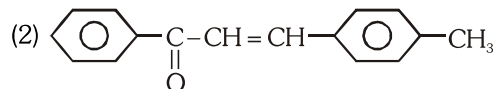
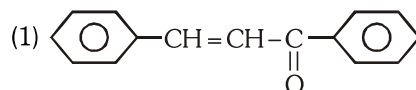
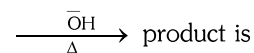
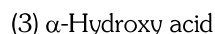
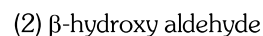
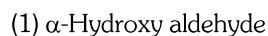
major product is



29. Which does not give Cu_2O with fehling solution



31. In aldol reaction of two aldehyde reaction product is



33. Give electrophilicity order of following carbonyl compounds

